Reducing 30-Day Diabetic Readmissions

Moving from reactive discharges to proactive, ML-driven care management.

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Diabetic readmissions are a costly, compounding crisis.

The Scope

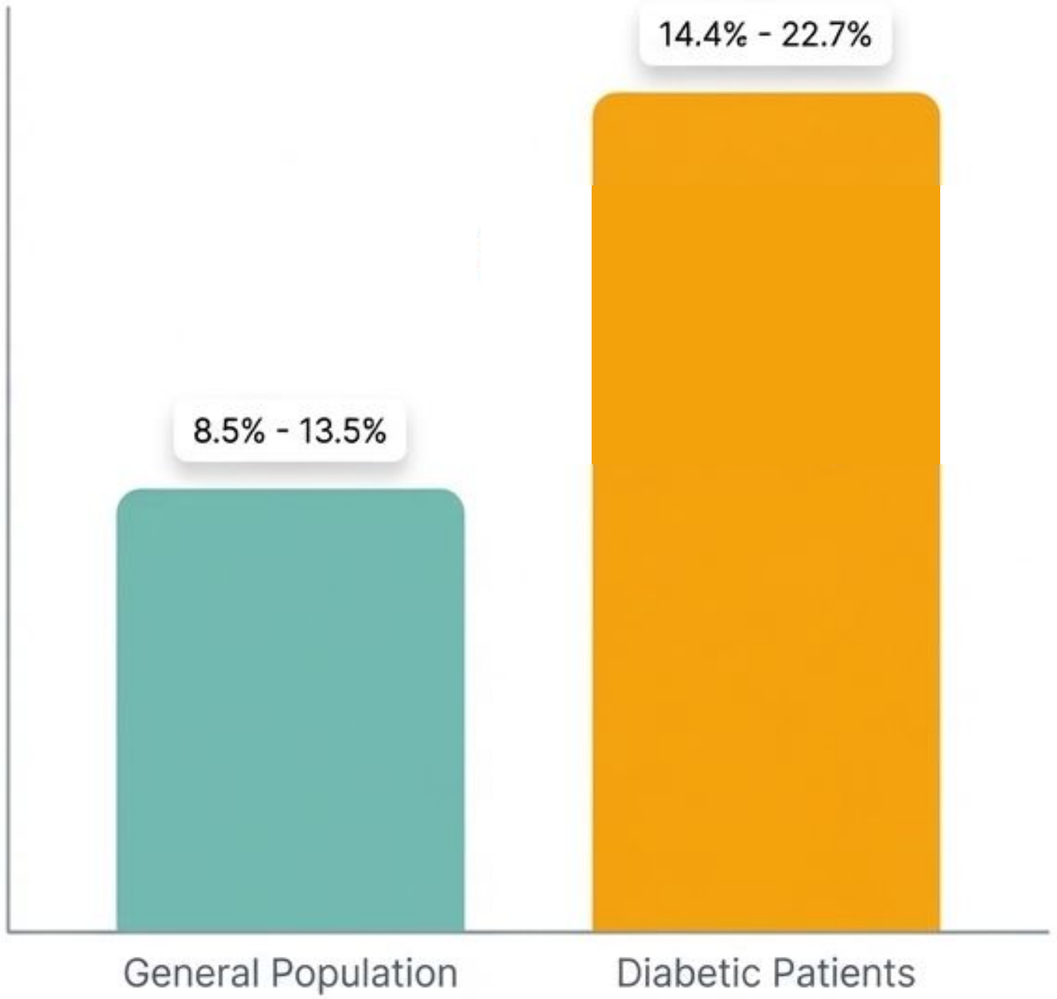
Diabetes impacts ~9.3% of the U.S. population, representing a massive chronic care burden.

The Disparity

Hospitalized diabetic patients face 30-day readmission rates nearly double the general hospital population.

The Impact

Avoidable readmissions drain hospital resources, trigger financial penalties, and severely degrade patient quality of life.



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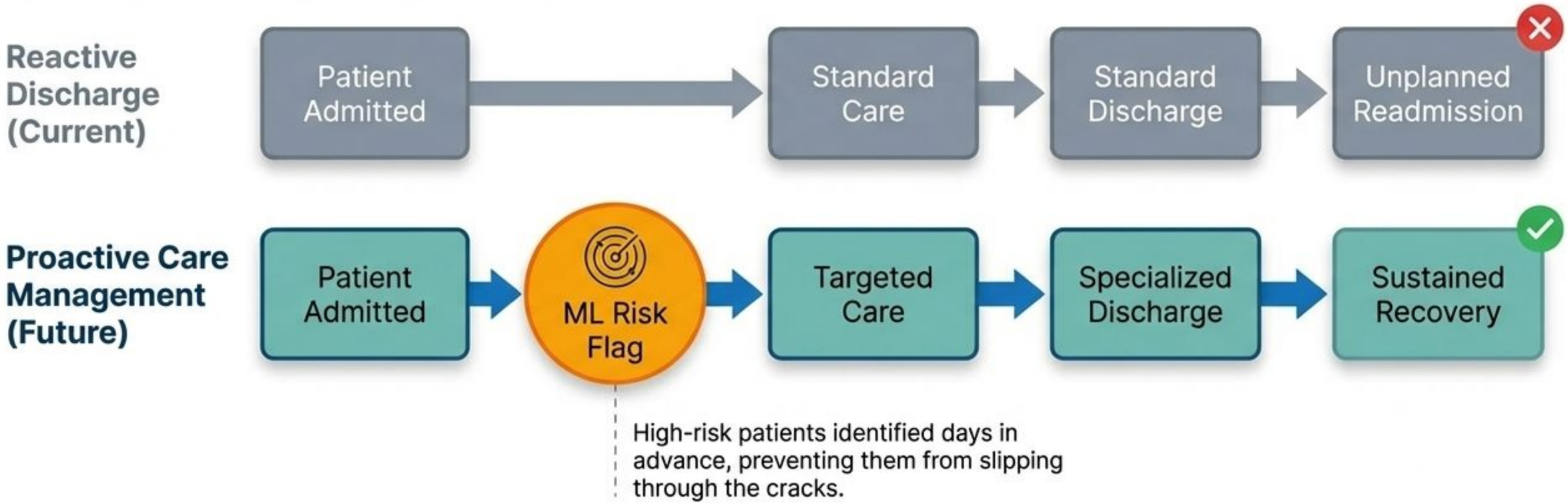
We can predict readmission risk before the patient leaves the hospital.

Predictive analytics acts as an early warning system, supporting discharge planners by mathematically identifying patients who need intensive follow-up before they are sent home.

Current vs. Future State



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A decision-support engine powered by historical hospital data

The Data

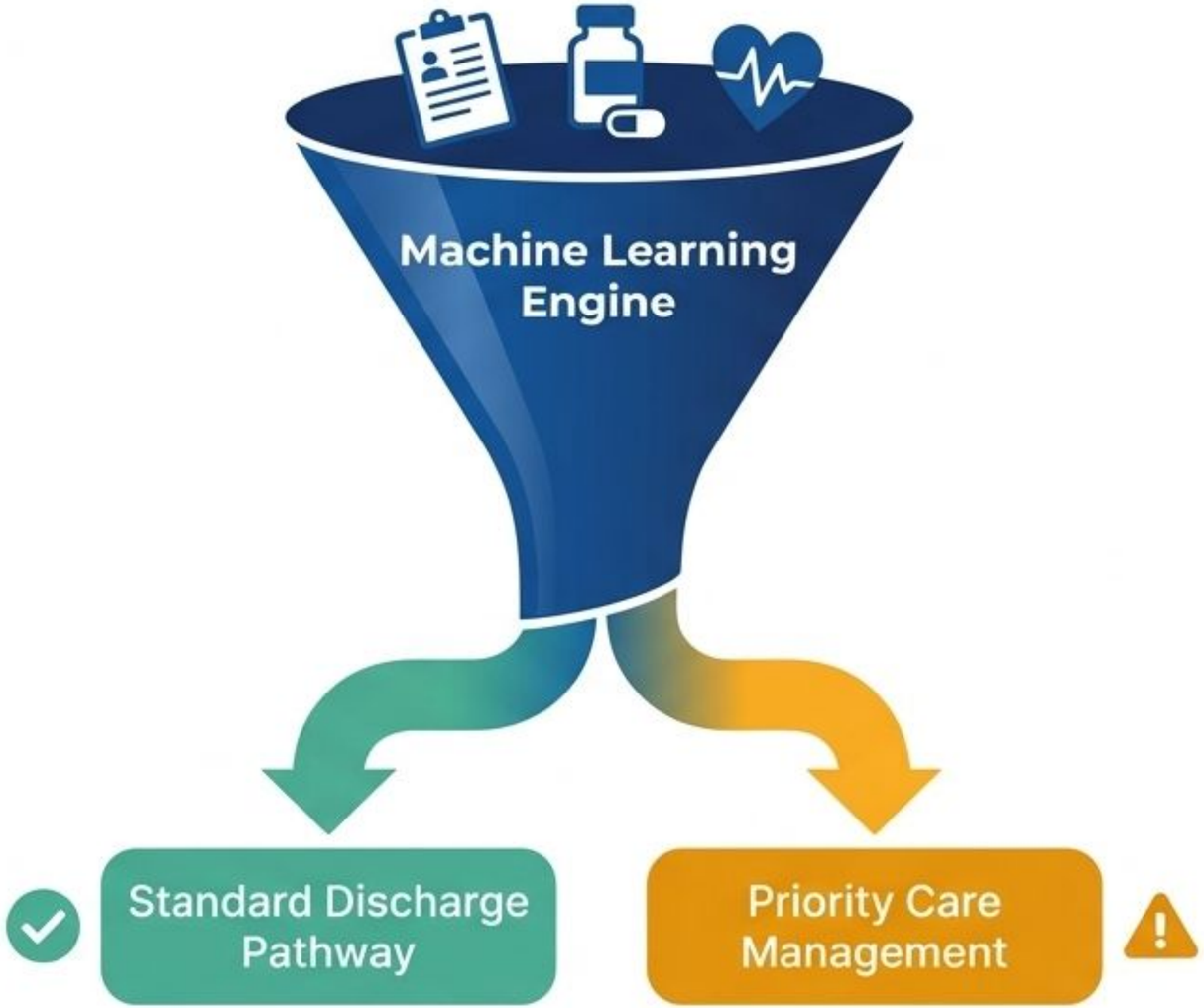
Analyzed 101,766 clinical encounters across 130 U.S. hospitals

The Inputs

Continuously reads 117 clinical EHR features (utilization history, diagnosis complexity, treatment intensity, demographics)

The Output

A simple, deployable binary flag identifying high-risk patients



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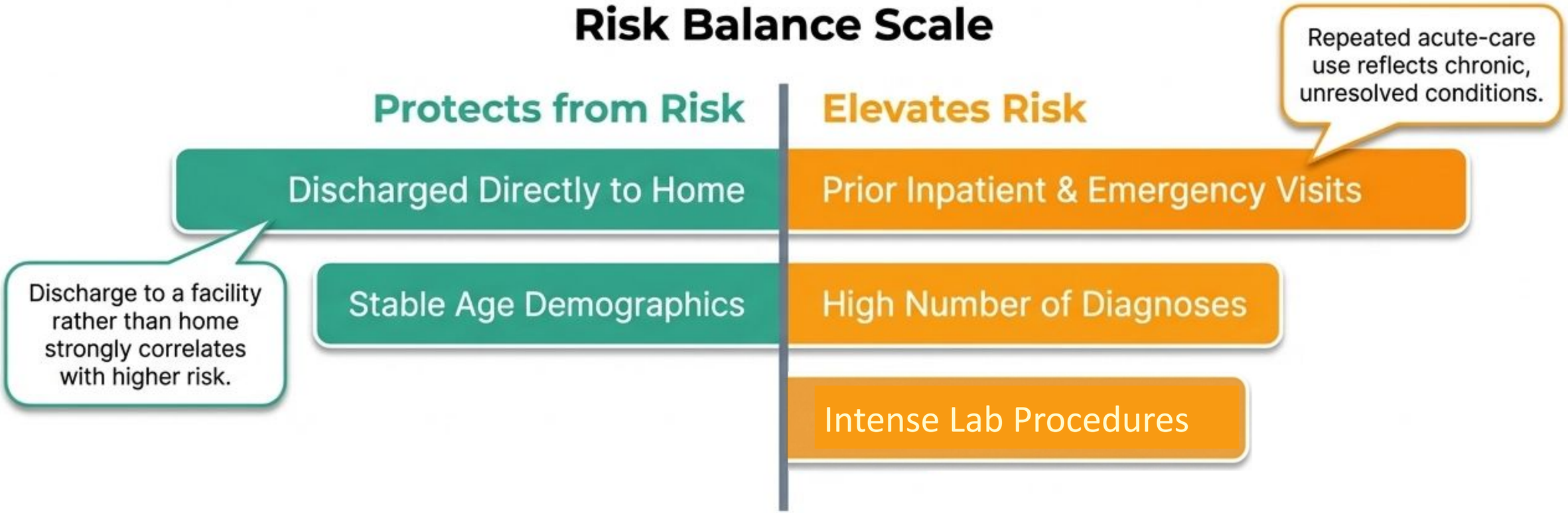
What actually drives a diabetic patient's risk of readmission?

The model confirms clinical intuition: past behavior predicts future behavior. Strongest predictors are known at the time of admission.

Risk Balance Scale



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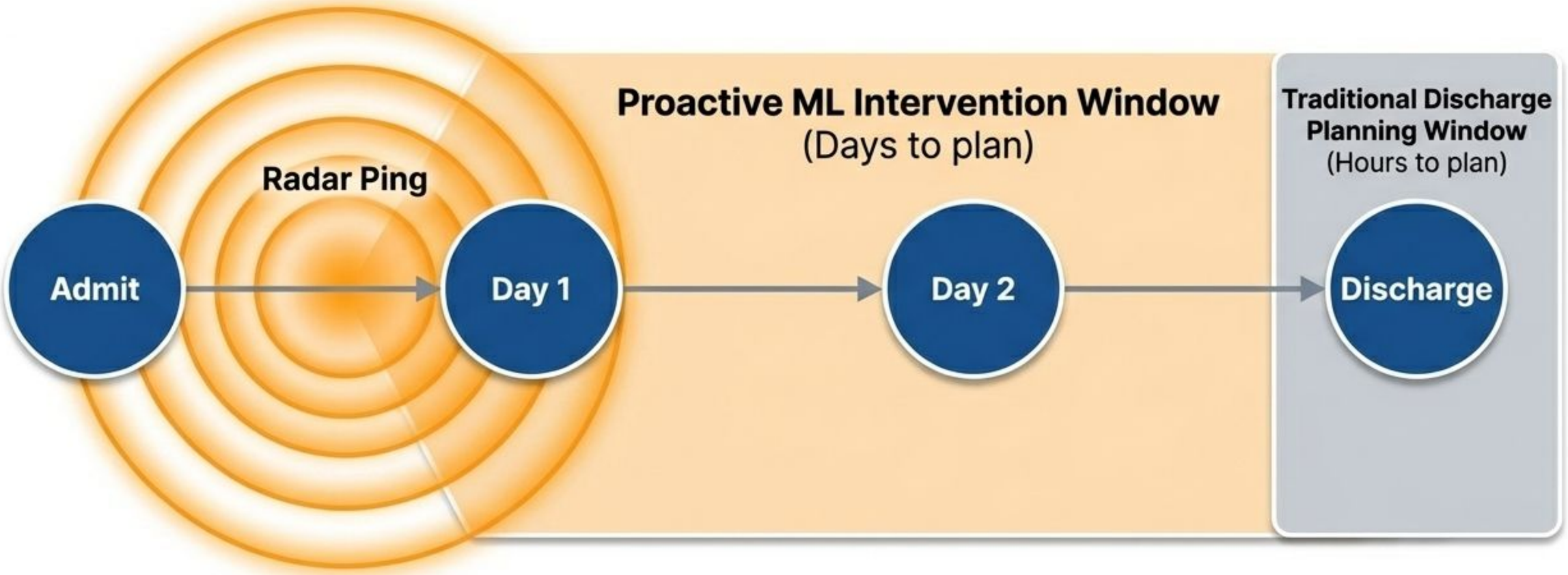
The strongest risk signals are visible on day one of admission.

Because the model relies heavily on historical utilization and demographic data, baseline risk is calculable the moment a patient enters the hospital. This gives staff days—not hours—to coordinate complex care.



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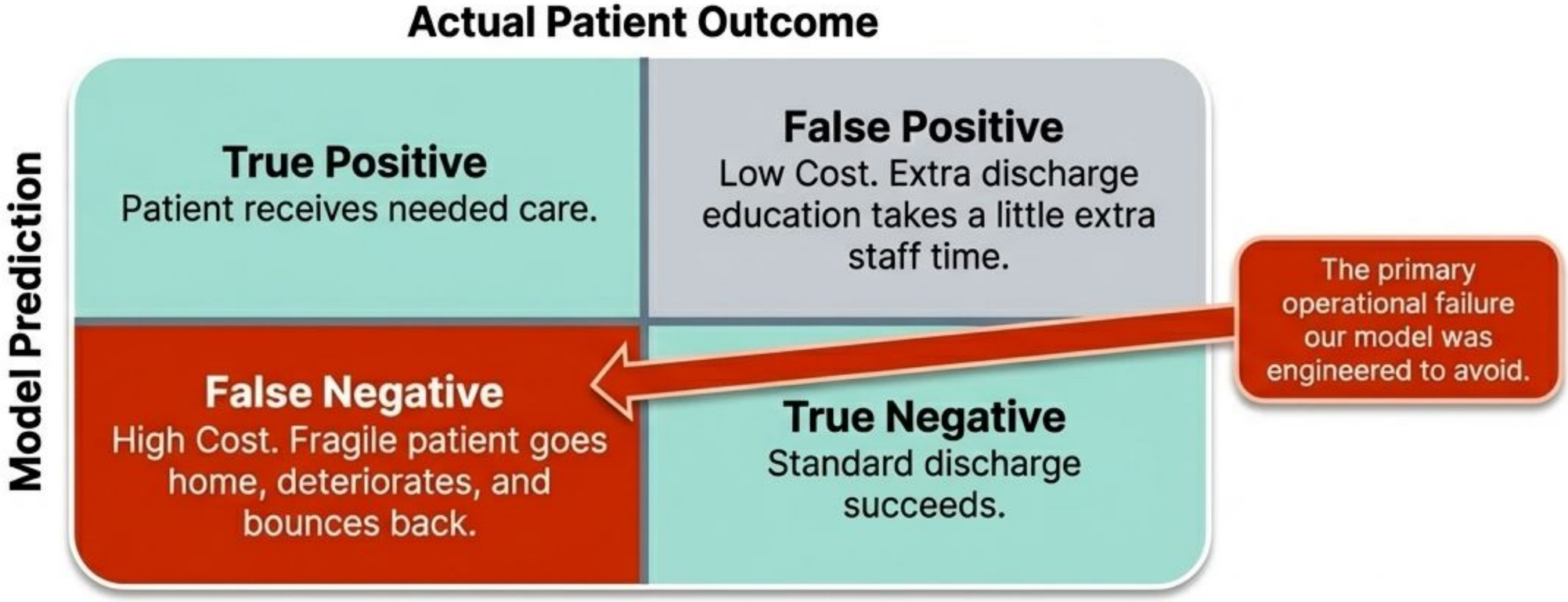
We tuned the model to cast a wider, safer net.

In healthcare, not all errors are equal. We optimized the model's threshold to capture the maximum number of true risks, intentionally prioritizing 'Recall' over perfect precision.

Asymmetric Cost Matrix



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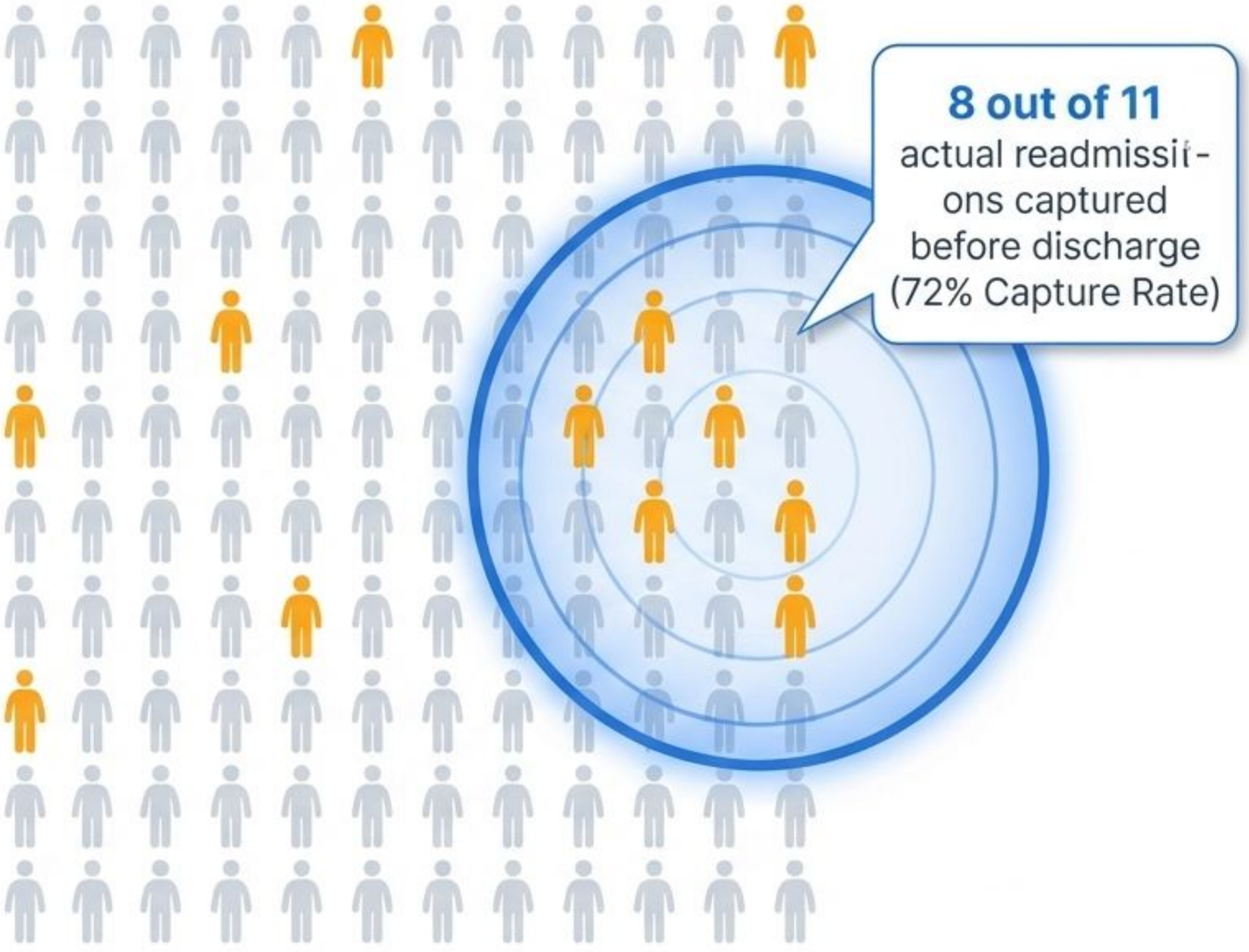
The clinical radar successfully captures 72% of true readmissions

At our selected operational threshold, the model catches ~72% of patients who will actually be readmitted within 30 days. By comparison, baseline random guessing catches roughly 4%.

The model splits the population, recommending half for standard discharge and focusing resources on the half that truly needs closer review.



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Transforming predictive math into operational ROI.

Earlier Intervention

Flagging patients immediately at admission provides care teams the critical lead time needed to coordinate specialized care, giving them days rather than hours.

Resource Allocation

Direct expensive, high-touch resources (home health visits, specialized pharmacy consults) only to the specific patient cohort mathematically proven to need them.

Cost Reduction

Preventing even a fraction of the 72% of captured readmissions will yield substantial financial savings and protect the hospital from government readmission penalties.

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Transparent limitations ensure responsible clinical deployment.

Decision Support, Not Replacement

The model is a “co-pilot” assistant. It flags mathematical risk, but clinical judgment remains the absolute final authority on discharge readiness.

Data Calibration

The current model relies on historical dataset patterns. It must be recalibrated against our hospital's modern EHR data to reflect current, localized treatment protocols.

Fairness & Equity

The model showed moderate performance disparities across some racial subgroups. Continuous monitoring is required to ensure interventions are distributed equitably to all.

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A phased approach to clinical integration

Phase 1: Silent Pilot
(Days 1–30)

- Run the model in the background on current live patient data.
- Validate accuracy without impacting actual clinical care.
- Calibrate the risk threshold.

Phase 2: Workflow Integration
(Days 30–60)

- Define specific clinical interventions for flagged patients.
- Implement mandatory medication reconciliation protocols.
- Train care coordinators on the dashboard.

Phase 3: Targeted Rollout
(Days 60+)

- Launch active dashboard in a single high-volume diabetic ward.
- Continuously monitor financial ROI.
- Track fairness metrics before hospital-wide expansion.

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